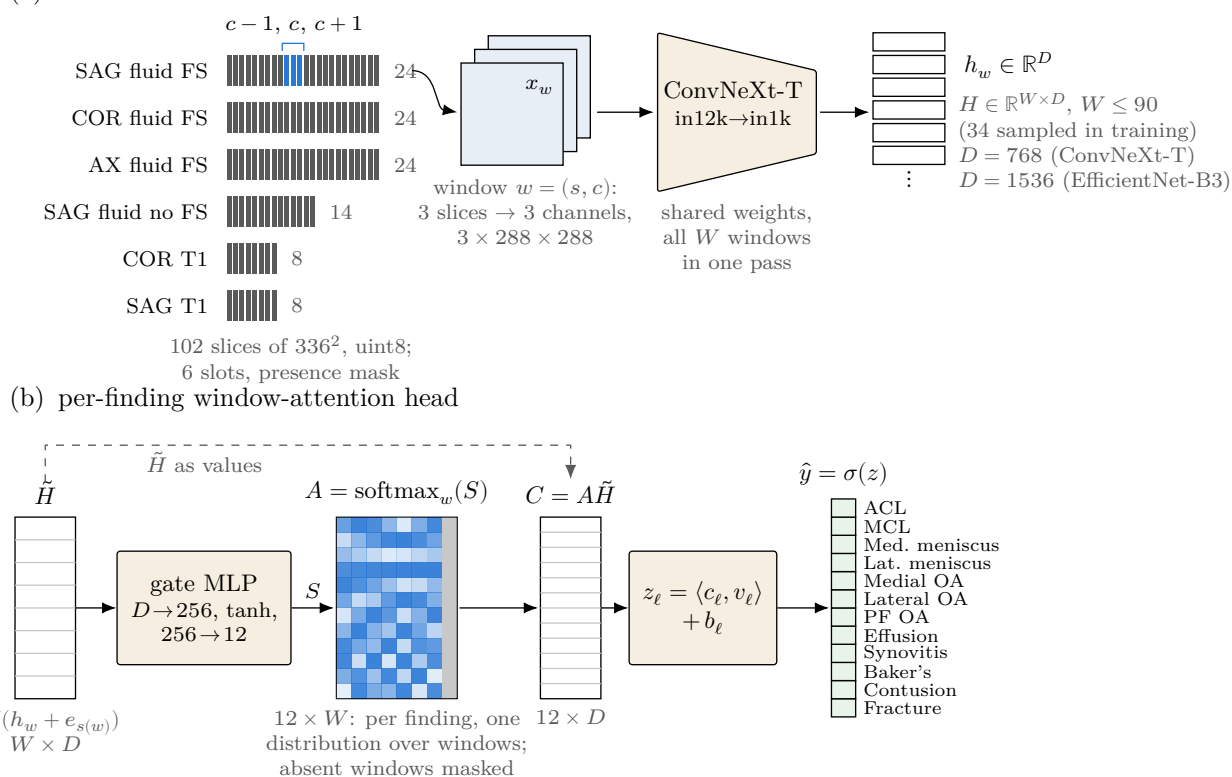
(a) window encoder
$c-1,\ c,\ c+1$
SAG fluid FS
COR fluid FS
AX fluid FS
SAG fluid no FS
COR T1
SAG T1
24
24
24
14
8
8
102 slices of 336^2, uint8;
6 slots, presence mask
x_w
window $w=(s,c)$:
3 slices → 3 channels,
$3 \times 288 \times 288$
ConvNeXt-T
in12k→in1k
shared weights,
all W windows
in one pass
$h_w \in \mathbb{R}^D$
$H \in \mathbb{R}^{W \times D}$, $W \le 90$
(34 sampled in training)
$D = 768$ (ConvNeXt-T)
$D = 1536$ (EfficientNet-B3)
(b) per-finding window-attention head
$\tilde{H}$ as values
$\tilde{H}$
$A = \mathrm{softmax}_w(S)$
$C = A\tilde{H}$
$\hat{y} = \sigma(z)$
gate MLP
$D \to 256$, tanh,
$256 \to 12$
S
$z_\ell = \langle c_\ell, v_\ell \rangle + b_\ell$
ACL
MCL
Med. meniscus
Lat. meniscus
Medial OA
Lateral OA
PF OA
Effusion
Synovitis
Baker's
Contusion
Fracture
$\mathrm{LN}(h_w + e_{s(w)})$
$W \times D$
$12 \times W$: per finding, one
distribution over windows;
absent windows masked
$12 \times D$